



BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Computer Science Department
Comp438 – QA course

Course Project

Research and Analysis of Recent Software Testing Topics

Testing for Accessibility (WCAG)

Students:

Mohammad Othman - 1221175

Farah Hammad - 1221176

Alaa Hammad - 1221177

Instructor: Faisal Shehadeh

Table of Contents

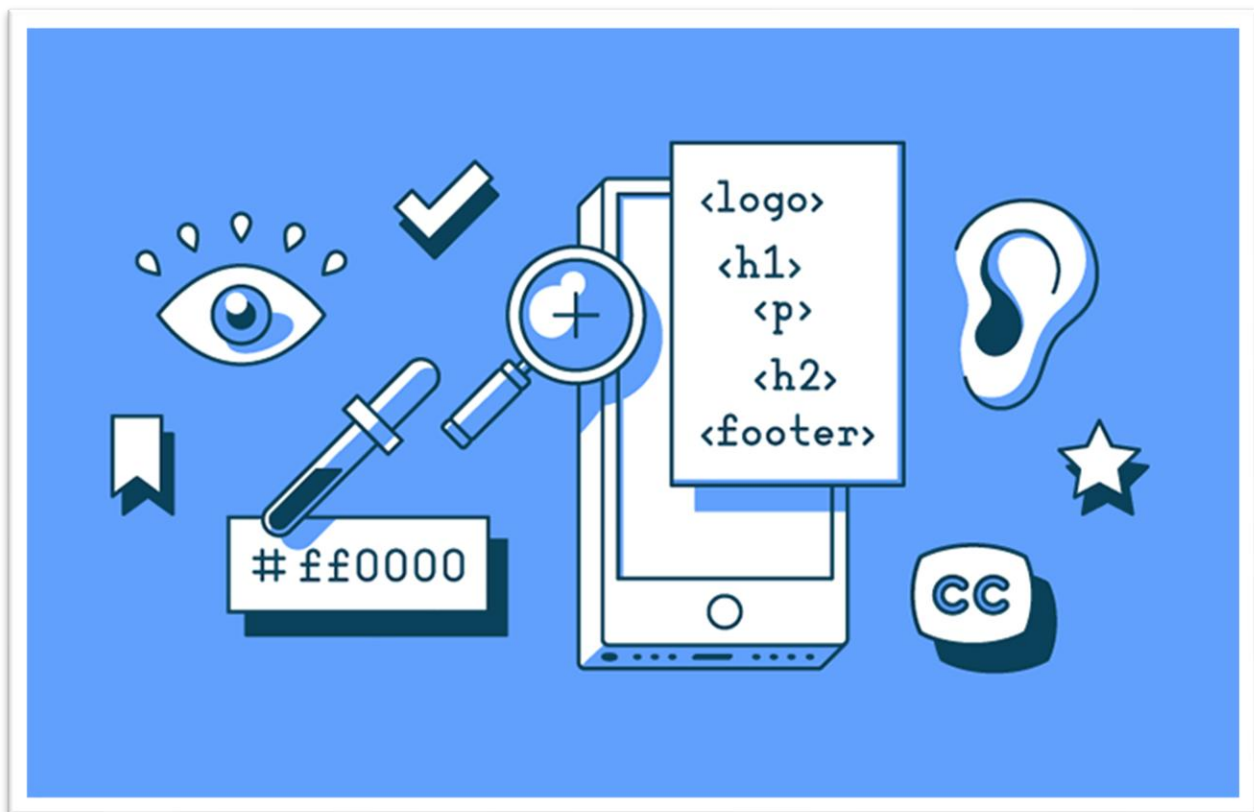
Introduction	3
Accessibility and SQA	6
Introduction to WCAG	7
WCAG Principles (POUR).....	8
WCAG Levels (A, AA, AAA)	9
Accessibility Testing Methodologies.....	10
Manual Testing:.....	10
Automated Testing:.....	10
Hybrid Approach	11
Accessibility Testing Tools.....	11
WAVE (WebAIM).....	11
Axe (Deque).....	12
Screen Readers (NVDA / VoiceOver)	12
Lighthouse (Google Chrome)	13
Challenges in Accessibility Testing	14
Benefits and Impact on Software Quality.....	14
Future Trends & Improvements.....	15
Conclusion	16
References.....	17

Introduction

❖ Accessibility in Programming

Accessibility in programming means making websites and applications easy to use for all people. This includes people with disabilities, but it also helps other users. For example, people who use mobile phones, slow internet, or small screens can benefit from accessible design. Accessibility helps users understand and use websites more easily.

Accessibility is about giving everyone the same chance to use digital services. People are different, but everyone has the same right to access information. Just like buildings have ramps and elevators for people in wheelchairs, websites should also support users with visual, hearing, or physical difficulties. Making websites accessible is the right thing to do.



In some countries, accessibility is required by law. When applications are accessible, more people can use them. This helps companies reach more users and improves the overall quality of the software.

❖ Who Needs Accessibility?

Many people need accessibility in digital applications. This includes people with permanent disabilities such as visual, hearing, physical, or learning difficulties. It also includes older people who may have problems with vision, hearing, or movement.

Accessibility is also important for people with temporary or situational difficulties. For example, a person with a broken arm, someone using a phone in bright sunlight, or a user in a noisy place. These situations make accessibility features helpful for everyone.

By supporting accessibility, applications become easier to use for all users and remove barriers for many people around the world.



❖ Why Should Applications Be Available for All?

Applications should be available for all people to ensure equal access. Everyone should be able to see, understand, and use digital content without difficulty. Accessibility helps users interact with applications in a simple and clear way.

Accessible design also improves usability for everyone. Features such as clear text, captions, and simple navigation make applications easier and more comfortable to use.

Finally, accessibility supports inclusion and fairness. Following accessibility standards such as **WCAG** help developers build better and more responsible software.

❖ Importance of Accessibility in Modern Applications

1. Daily use of applications

Many people use mobile apps and websites every day for communication, learning, shopping, and work. Applications should be easy for everyone to use.

2. Easy access for all users

When an application is accessible, more people can get information and services without problems.

3. Problems without accessibility

If accessibility is not considered, some users cannot use the application. This includes people with visual, hearing, or physical disabilities.

4. Large number of affected users

According to World Health Organization, more than one billion people worldwide live with some form of disability. This shows why accessibility is very important.

5. Ethical responsibility

Accessibility is about fairness and equal rights. Developers should design applications that include all users.

6. Standards and guidelines

Organizations such as the World Wide Web Consortium provide accessibility guidelines to help create better and fairer digital applications.

Accessibility and SQA

Accessibility is an important part of software quality. A good application should work correctly and be easy to use for all users. When accessibility is included, the software becomes more useful and fairer for everyone. Software testing helps check accessibility by finding problems such as small text, unclear buttons, or missing captions. Finding these problems early helps improve the application before users use it.

Accessibility testing also improves the user's experience. When applications are simple to read and easy to navigate, users feel more comfortable and less confused. This helps them finish tasks faster and with less effort. Accessibility is a key part of Quality Assurance because it ensures the application meets quality standards and user needs.

By testing accessibility, developers can reduce errors, remove barriers, and deliver a more reliable and high-quality application.

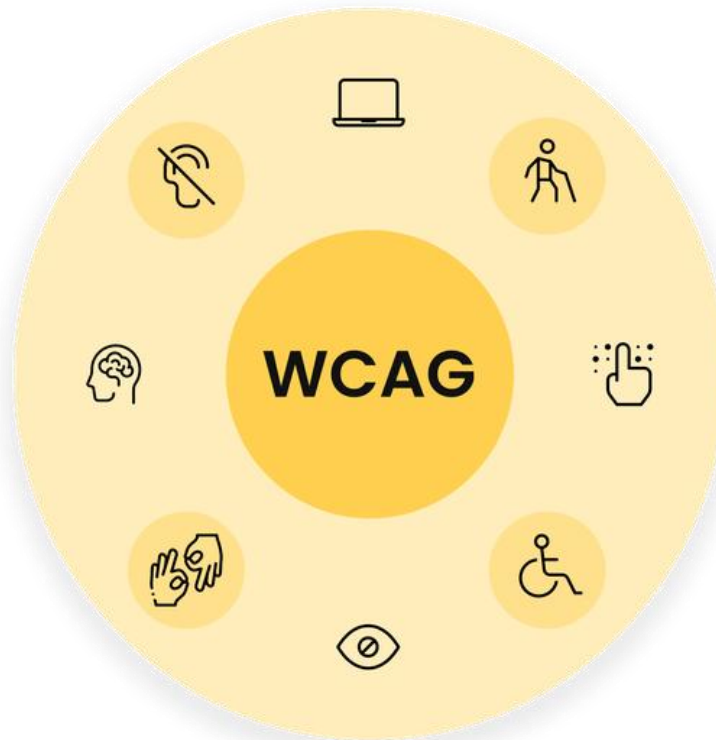


Introduction to WCAG

WCAG stands for Web Content Accessibility Guidelines. These guidelines are international standards that explain how to make websites and applications accessible for all users. The main goal of WCAG is to help people with disabilities use digital content more easily and without barriers.

WCAG guidelines are created by an international organization called the World Wide Web Consortium. This organization works on setting global web standards to improve the quality and accessibility of the internet. WCAG provides clear rules that developers and testers can follow when building digital applications.

Today, WCAG is widely used by developers, software testers, and large companies around the world. It is considered the main reference for accessibility testing. By following WCAG guidelines, teams can ensure their applications are accessible, fair, and usable for everyone.



WCAG Principles (POUR)

WCAG guidelines are based on four main principles called POUR. These principles explain how digital content should be designed to be accessible to all users.

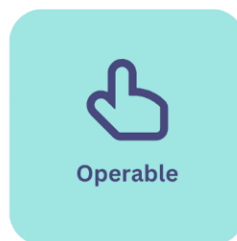
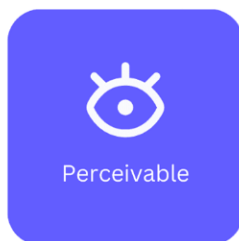
Perceivable means that users must be able to see or hear the content. For example, images should have text descriptions, and videos should include captions so users can understand the information.

Operable means that users must be able to use the application easily. All functions should work with a keyboard, and users should have enough time to read and interact with content.

Understandable means that the content should be clear and easy to understand. Text should use simple language, and the application should behave in a predictable way.

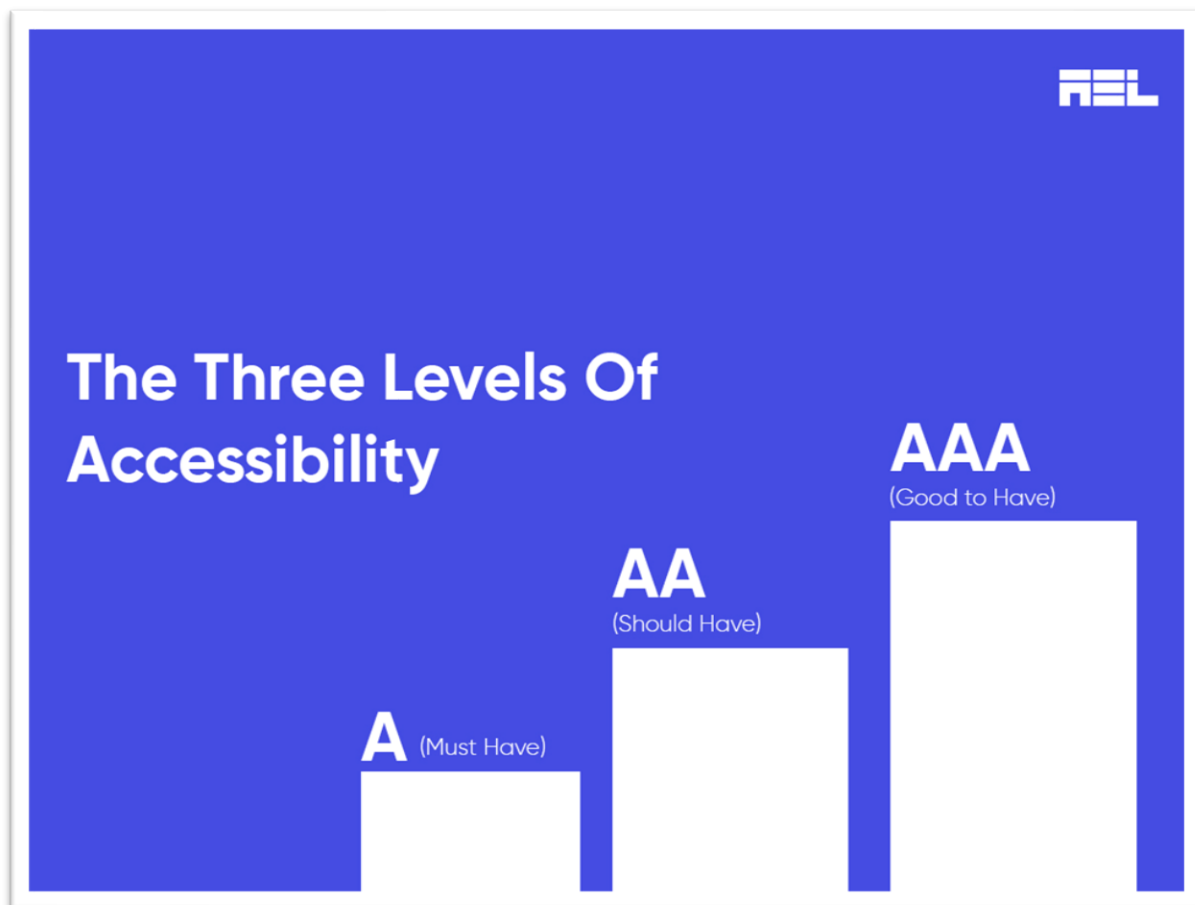
Robust means that the content should work with different devices and assistive technologies. Applications should be compatible with screen readers and other tools used by people with disabilities.

POUR Principles



WCAG Levels (A, AA, AAA)

1. **Level A** is the basic level of accessibility. It fixes the most serious problems that prevent users from accessing content.
2. **Level AA** is the most common level used by organizations. It provides a good level of accessibility and is often required by laws and policies.
3. **Level AAA** is the highest level of accessibility. It offers the best user experience, but it is difficult to achieve for all content.



Accessibility Testing Methodologies

Manual Testing:

Involves human testers interacting with content as end users (e.g. using only a keyboard or a screen reader). Testers follow defined scenarios (such as “no sight, no sound, no mouse”) to simulate disabilities. This typically means using assistive technology (NVDA, VoiceOver, etc.) and checking the site’s functionality and content manually. Manual testing finds issues that automated tools cannot, such as whether alt text is meaningful, whether form workflows make sense, or how on-focus events behave in practice. In fact, test-party research notes that automated scans catch only about 30–40% of accessibility problems; the rest (semantics, context, usability) manifest only under real assistive technology use. Benefits of manual testing include human judgment of content and usability (ensuring WCAG criteria like understandable text and logical flow). Limitations are that it is time-consuming, resource-intensive, and can be inconsistent across testers. Expert surveys emphasize that “only a human can determine true accessibility”, so manual audits remain essential for complex WCAG checks (for example, confirming keyboard navigation is logical and comprehensive).

Automated Testing:

Uses software tools to scan code and flag obvious WCAG violations. These tools parse HTML/CSS/ARIA to detect missing alt attributes, color-contrast failures, broken headings or labels, invalid ARIA, and other programmatically-detectable issues. Automated tests run extremely fast: they can audit thousands of pages in minutes and run in CI/CD pipelines to catch regressions. For example, popular tools like Axe (Deque) and WAVE (WebAIM) provide browser extensions or APIs to generate issue reports automatically. However, automated scanning has well-known gaps. Research shows most tools cover only about one-sixth of WCAG success criteria, catching roughly 30–40% of actual accessibility issues. They excel at *consistency* (they never tire or overlook rule-checks) but cannot evaluate meaning or usability. For instance, tools can detect that an image has an alt attribute, but not whether the alt text aptly describes the image (semantic accuracy). Automated tools also miss context-sensitive problems: they can’t judge if a link label makes sense out of context, if focus order is intuitive, or if error messages are helpful. WCAG checkpoints involving user interaction (like: form error recovery, audio transcription quality, or logical reading order) typically evade automation. In short, automated tests provide a quick baseline by detecting syntactic or binary violations (missing attributes, contrast ratio failures, heading hierarchy, and other.), but leave most subjective or dynamic checks to humans

Hybrid Approach

Combining both methods is best practice. Most experts recommend using automated scans to cover broad sweeps and then following up with targeted manual or assistive-technology testing to catch what machines miss. In a hybrid workflow, developers run automated audits as the first pass (fast checks of code-level issues), and then human testers or QA specialists perform deeper reviews of key pages and user pathways. This ensures *complete* coverage: automation handles repetitive, high-volume checks, while manual assessment verifies semantic relevance, keyboard operability, and end-user experience. The WAVE tool itself explicitly notes that it “cannot tell you if your web content is accessible – only a human can determine true accessibility”. Thus, a blended strategy (often called “automated+manual” testing) leverages the strengths of both. Automated tools integrated into CI/CD pipelines catch many regressions early, and periodic manual audits (including screen reader testing) confirm compliance with the remaining WCAG criteria. This 360° approach is the most reliable way to ensure WCAG compliance in practice.



Accessibility Testing Tools

WAVE (WebAIM)

WAVE is a free accessibility evaluation tool developed by WebAIM. It visually highlights accessibility issues directly on the web page using icons and indicators. WAVE helps identify problems such as missing alt text, contrast issues, and structural errors.

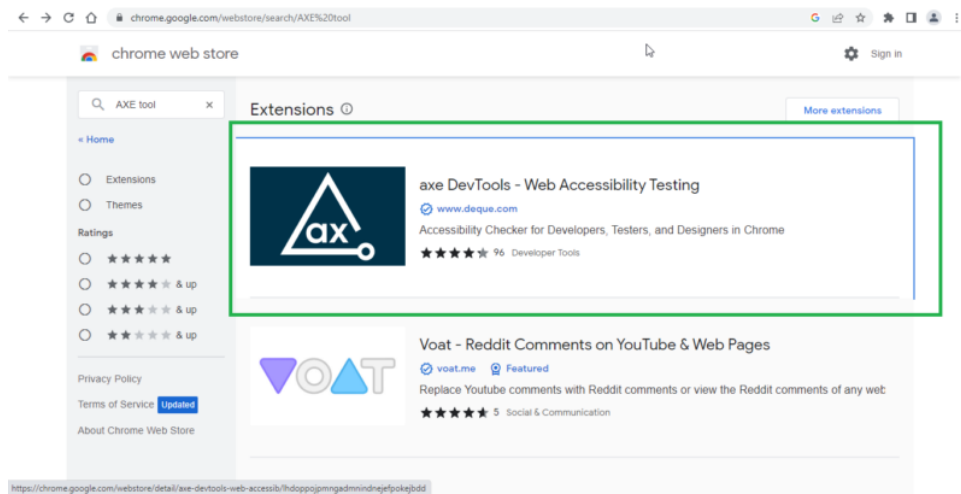
Its main strength is that it supports human evaluation by clearly showing where problems exist and explaining them. However, WAVE cannot automatically test large websites and does not replace manual testing. WebAIM itself states that WAVE is meant to assist human judgment, not replace it.

Axe (Deque)

Axe is a popular open-source accessibility testing tool developed by Deque Systems. It offers browser extensions and integrates with testing frameworks such as Cypress and Playwright. Axe is known for accurate results and minimal false positives.

It provides detailed reports with explanations and code-level guidance, making it useful

for developers. Like other automated tools, Axe cannot assess content quality or usability, but it is highly effective for detecting programmatic WCAG violations and preventing regressions during development



Screen Readers (NVDA / VoiceOver)

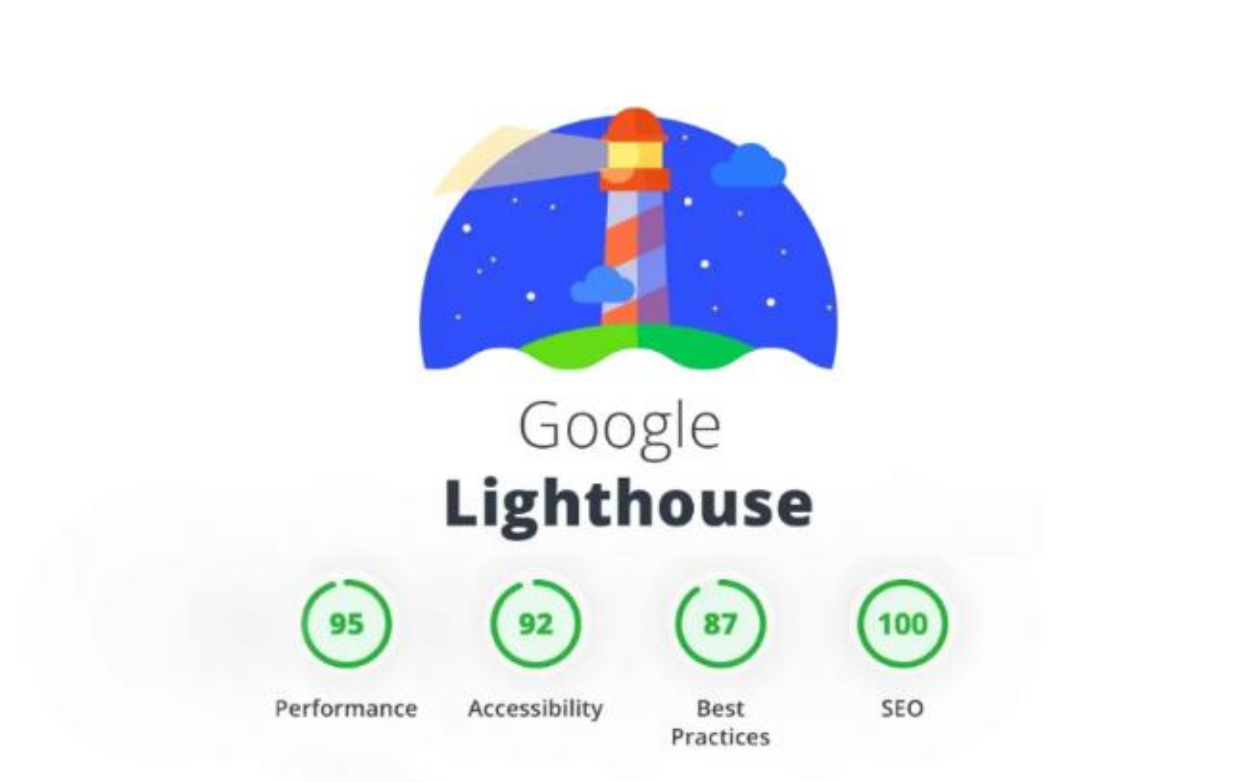
Screen readers are essential tools for accessibility testing. NVDA (Windows) and VoiceOver (macOS and iOS) allow testers to experience websites as visually impaired users do. These tools read page content aloud and announce buttons, links, headings, and form fields.

Testing with screen readers reveals issues such as missing labels, incorrect reading order, or unclear navigation that automated tools often miss. Because screen readers rely heavily on proper HTML structure and semantics, they are critical for verifying real WCAG compliance. Using screen readers during testing ensures that accessibility requirements are met from a real user perspective.

Lighthouse (Google Chrome)

Lighthouse is an automated auditing tool built into Chrome DevTools. It provides an accessibility score and highlights issues related to WCAG guidelines. Lighthouse is easy to use and requires no setup, making it suitable for quick checks.

However, Lighthouse tests only one page at a time and does not cover all accessibility rules. Its score should be used as an indicator, not proof of full compliance. Many teams use Lighthouse alongside more specialized tools such as Axe.



Challenges in Accessibility Testing

One challenge in accessibility testing is that many accessibility rules need human testing. Some requirements cannot be checked by tools and need people to review and test them. This is difficult for large websites and for websites that change often.

Another challenge is testing large and dynamic websites. These websites have many pages and different versions of content. Because the content changes frequently, it is hard to check all pages and ensure they follow accessibility rules.

A third challenge is the use of third-party content. On many websites, content is added or changed by external users or services. This makes it harder to control accessibility and always keep the website accessible.

Finally, accessibility standards are mainly designed for web pages. Applying these standards to non-web systems, such as other information and communication technologies, can be difficult and unclear.

Benefits and Impact on Software Quality

Accessibility testing improves the overall quality of software. When applications are accessible, they are easier to use for all users, not only people with disabilities. Clear text, simple navigation, and well-labeled buttons help users complete tasks with less confusion and fewer errors.

Accessibility also improves the user experience. When users can understand and interact with the application easily, they feel more comfortable and satisfied. This reduces frustration and increases trust in the software. As a result, accessible applications are often more successful and widely used.

In addition, accessibility testing helps developers build more reliable and well-structured applications. Following accessibility guidelines encourage clean design, consistent layouts, and correct use of code. This makes the software easier to maintain and improve over time.

Accessibility also has a positive impact on organizations. Accessible software can reach more users, including older adults and people with disabilities. According to the World Health Organization, more than one billion people worldwide live with some form of disability. By supporting accessibility, companies can serve a larger audience and meet legal and ethical requirements.

Finally, accessibility testing supports fairness and inclusion. International standards such as WCAG, developed by the World Wide Web Consortium, help teams build software that is usable and fair for everyone. This makes accessibility an important part of software quality assurance.

Future Trends and Improvements

In the future, accessibility testing will become more important in software development. More teams will include accessibility from the beginning of the project instead of testing it at the end. This helps reduce problems and saves time and money.

New tools are also improving accessibility testing. Some tools use automation and artificial intelligence to find accessibility issues faster. These tools can help testers detect problems early and suggest simple fixes, but human testing will still be important.

Another improvement in future is better awareness and training. More developers and testers are learning about accessibility and WCAG guidelines. This helps teams design applications that are accessible from the start.

Finally, laws and policies related to accessibility are increasing in many countries. Because of this, organizations will focus more on building accessible and inclusive applications. Accessibility will become a standard part of software quality assurance.



Conclusion

Accessibility testing is an important part of software quality assurance. It helps ensure that websites and applications are easy to use for all people, including users with disabilities. By following accessibility guidelines, developers can remove barriers and create more inclusive digital products.

Although accessibility testing has some challenges, such as time, cost, and the need for human testing, its benefits are greater. Accessible applications improve user experience, reduce errors, and increase software quality. They also help organizations reach more users and meet legal and ethical requirements.

Standards such as WCAG provide clear guidance for testing accessibility. Developed by the World Wide Web Consortium, these guidelines help teams build software that is fair, usable, and reliable. For these reasons, accessibility testing should be standard practice in modern software development.

References

- [1]. *Web Content Accessibility Guidelines (WCAG) 2.1. W3.org.*
- [2]. *What is accessibility? MDN Web Docs.*
- [3]. *Why Digital Accessibility Benefits Everyone. APHSA.*
- [4]. *Automated Accessibility Testing: What It Catches and What It Misses*
- [5]. *WAVE Web Accessibility Evaluation Tools*
- [6]. *Google Lighthouse Review | Automated Accessibility Testing Tools*
- [7]. *Using the NVDA Screen Reader to Test Web Accessibility*
- [8]. *Screen Reader Testing Guide: A Developer's Complete Methodology*
- [9]. *WebAIM: Screen Reader User Survey #10 Results*
- [10]. *World Wide Web Consortium. (n.d.). Accessibility conformance challenges.*